

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

**ENVIRONMENTAL SCIENCE****Paper 1 (Generated practice paper - Set 13)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

**Time allowed: 3 hours****Materials**

For this paper you may use:

- a calculator.

**Instructions**

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

**Information**

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

**For Examiner's Use**

| Question | Mark |
|----------|------|
| 1        |      |
| 2        |      |
| 3        |      |
| 4        |      |
| 5        |      |
| 6        |      |
| 7        |      |
| 8        |      |
| 9        |      |
| 10       |      |
| 11       |      |
| TOTAL    |      |

**Answer all questions in the spaces provided.**

**0 1**

The Universal Soil Loss Equation (USLE) is  $A = R \times K \times LS \times C \times P$ . Table 1 gives factor values for a farm.

**Table 1**

| Factor                  | Description                                                                                         | Value        |
|-------------------------|-----------------------------------------------------------------------------------------------------|--------------|
| R                       | Rainfall erosivity for the farm                                                                     | 110          |
| K                       | Soil erodibility: loam                                                                              | 0.67         |
| LS                      | Slope length and gradient: 120 m, 6%                                                                | 1.35         |
| C (crop type x tillage) | Beans (0.50) with spring ploughing (0.90); or wheat (0.35) with mulch tillage (0.60)                | see question |
| P                       | Ploughing up and down slope 1.0; contour ploughing 0.50; strip cropping with contour ploughing 0.25 | see question |

**Table 2**

| Soil erosion class | Rate of soil loss / $\text{t ha}^{-1} \text{ yr}^{-1}$ |
|--------------------|--------------------------------------------------------|
| Low                | under 5                                                |
| Moderate           | 5-10                                                   |
| High               | 10-20                                                  |
| Very high          | 20-30                                                  |
| Severe             | over 30                                                |

**0 1**

**1**

Calculate the annual soil loss for beans with spring ploughing, ploughed up and down the slope. Give your answer to one decimal place.

**[2 marks]**

**Answer**

t/ha/yr

**0 1**

**2**

Calculate the annual soil loss for wheat with mulch tillage and strip cropping with contour ploughing.

**[2 marks]**

Answer  t/ha/yr

**0 1 . 3** The natural rate of soil formation on the farm is  $1 \text{ t ha}^{-1} \text{ yr}^{-1}$ . Use your answers and Table 2 to evaluate the sustainability of the two farming systems. **[3 marks]**

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**0 1 . 4** Explain why the USLE gives only an estimate of soil loss. **[2 marks]**

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**0 1 . 5** Suggest which single factor in the USLE a farmer can change most easily. **[1 mark]**

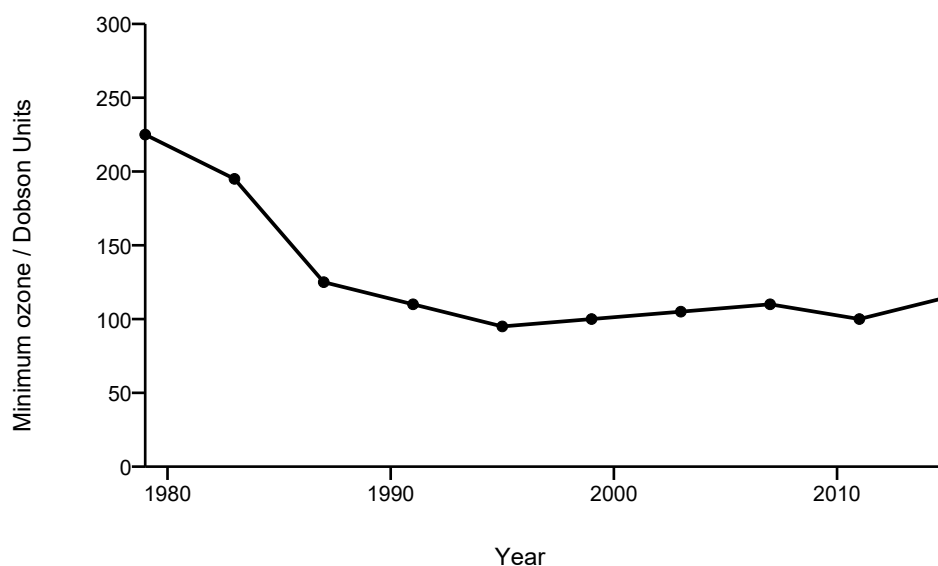
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**10**

**0 2**

Figure 1 shows the minimum ozone concentration recorded over Antarctica each spring between 1979 and 2015.

**Figure 1****0 2****1**

Describe the trend shown in Figure 1.

**[2 marks]****0 2****2**

Calculate the percentage decrease in minimum ozone between 1979 and 1995.

**[2 marks]****Answer**

**%****0 2****3**

Explain why ozone depletion is most severe over Antarctica in spring.

**[4 marks]**

0 2 . 4

Explain why the data in Figure 1 alone could not confirm that CFCs were the cause of the depletion, and describe how the cause was confirmed.

[2 marks]

10

0 3 . 1

Explain why nitrates in drinking water can cause **blue baby syndrome** (methaemoglobinaemia).

[3 marks]

0 3 . 2

Explain why the link between nitrates in drinking water and stomach cancer is uncertain.

[2 marks]

0 3 . 3

Describe **three** ways farmers in a Nitrate Vulnerable Zone can reduce nitrate leaching into aquifers.

[3 marks]

1 .....

2 .....

3 .....

0 3 . 4

Explain how phosphates are removed from sewage effluent and from lakes polluted in the past.

[2 marks]

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0 4 . 1

Describe how bioleaching is used to extract copper from low-grade ores or spoil heaps, and explain why it may be more sustainable than smelting.

[5 marks]

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04 . 2

Describe the process of phytomining.

[3 marks]

04 . 3

Explain how polymer adsorption may provide a low-energy source of uranium in the future.

[2 marks]

10

05 . 1

Explain how the control of emission **location** and emission **timing** can reduce the severity of pollution. Use examples.

[6 marks]

**Emission location** .....

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**Emission timing** .....

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**0 5 . 2**

Explain the **polluter pays** principle and suggest why it encourages pollution prevention.

**[2 marks]**

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**0 5 . 3**

Explain the **precautionary principle** as applied to the release of a new chemical.

**[2 marks]**

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**10**

06

A student estimated the population of ground beetles in a field using the Lincoln Index. 60 beetles were caught in pitfall traps, marked with a spot of paint and released. Three days later 45 beetles were caught, of which 9 were marked.

06

1 Calculate the estimated population size.

[2 marks]

Answer

beetles

06

2 State **four** assumptions of the Lincoln Index method.

[4 marks]

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06

3 Explain how the estimate would be affected if the paint made the beetles more likely to be eaten by birds.

[2 marks]

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06

4 Suggest why the second sample was taken after three days rather than the next day or after a month.

[2 marks]

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10

07 Mercury released into a lake from a factory was in the inorganic form. Fish in the lake were later found to contain high concentrations of methyl mercury, an organic form.

07 . 1 Suggest how the inorganic mercury was converted to methyl mercury.  
[2 marks]

07 . 2 Explain why methyl mercury is a more hazardous pollutant than inorganic mercury.  
[3 marks]

07 . 3 Explain why the health risk from eating fish from the lake may remain for many years after the discharges stop.  
[2 marks]

07 . 4 Suggest **three** factors that should be included in a Critical Pathway Analysis of a proposed mercury discharge.  
[3 marks]

1 .....

2 .....

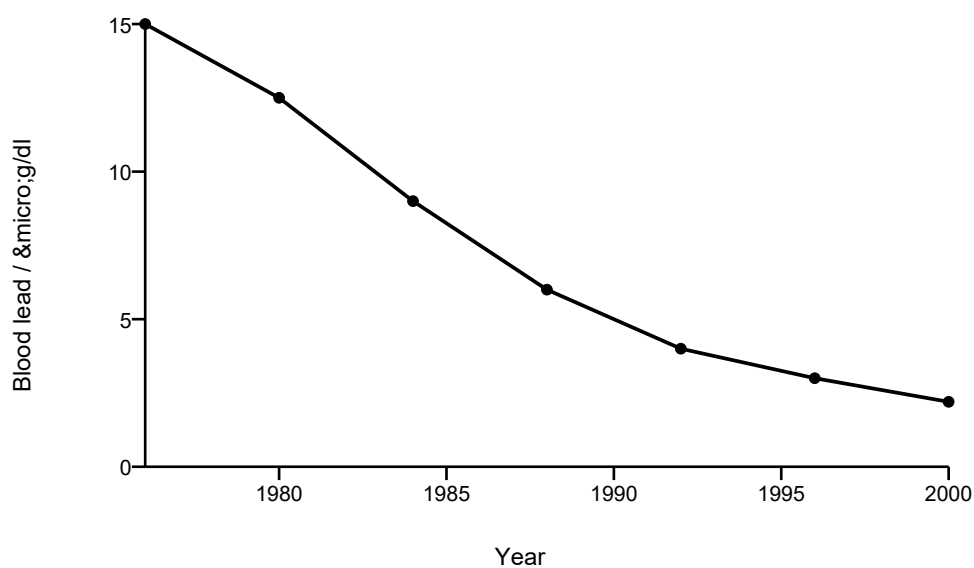
3 .....

10

0 8

Figure 1 shows the mean concentration of lead in the blood of children in a large city between 1976 and 2000, and the year in which lead additives in petrol began to be phased out.

Figure 1



0 8

1 Calculate the percentage decrease in mean blood lead between 1976 and 2000.

[2 marks]

Answer

%

08 . 2

Explain why tetraethyl lead was added to petrol and why this caused a public health threat.

[3 marks]

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08 . 3

Suggest **two** reasons, other than petrol additives, why blood lead concentrations continued to fall after 1990.

[2 marks]

1 .....

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2 .....

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08 . 4

Explain how adding small amounts of phosphoric acid to a water supply reduces lead in drinking water.

[2 marks]

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08 . 5

Explain why discarded lead fishing weights and shotgun pellets can poison waterfowl.

[1 mark]

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10

09 . 1

Which of the following properties would make a pesticide **least** likely to biomagnify? Tick one box.

[1 mark]

- A High liposolubility and high persistence
- B Low liposolubility and low persistence
- C High liposolubility and low persistence
- D Low liposolubility and high persistence

☐☐☐☐

09 . 2

Explain the difference between **bioaccumulation** and **biomagnification**.

[2 marks]

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09 . 3

Explain why pesticides are described as an unusual type of pollutant.

[2 marks]

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5

10 . 1

Explain why the development of energy storage technologies is essential if renewable energy resources are to provide most of a country's electricity.

[9 marks]

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Extra space

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**10** . **2** State what is meant by **peak shaving**.

**[1 mark]**

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**10**

Do not write outside the box

**Write an essay on one of the following topics.**

|   |   |   |   |
|---|---|---|---|
| 1 | 1 | . | 1 |
|---|---|---|---|

Discuss how knowledge of the properties of pesticides can be used to reduce their environmental impacts while maintaining food production.

**[25 marks]**

**OR**

**1 1 . 2**

Discuss how knowledge of the properties of pollutants can be used to reduce the severity of their impact on the environment.

**[25 marks]**

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There is no text or other markings on the paper.

[illegible]

Do not write outside the box

A large rectangular box containing 30 horizontal dotted lines for writing.

**END OF QUESTIONS**